HIDDEN BARGAINS

FINDING UNDERVALUED PLAYERS







NOVA IMS
INFORMATION MANAGEMENT SCHOOL

MOTIVATION

MODERN TRANSFER MARKET IS UNPREDICTABLE



LET'S STOP GUESSING. LET'S MEASURE PERFORMANCE OBJECTIVELY AND FIND PLAYERS WHO ARE ACTUALLY GOOD AND CHEAP.

TWO WORLDS OF FOOTBALL





















KICKING OFF:
DATA PREPARATION

PLAYER SELECTION



MISSING VALUES

?

THE DNA OF A FORWARD:
OUR PERFORMANCE VARIABLES

PROGRESSION

SCORING

PRESSURE

CHANCE CREATION

TECHNICAL ERRORS

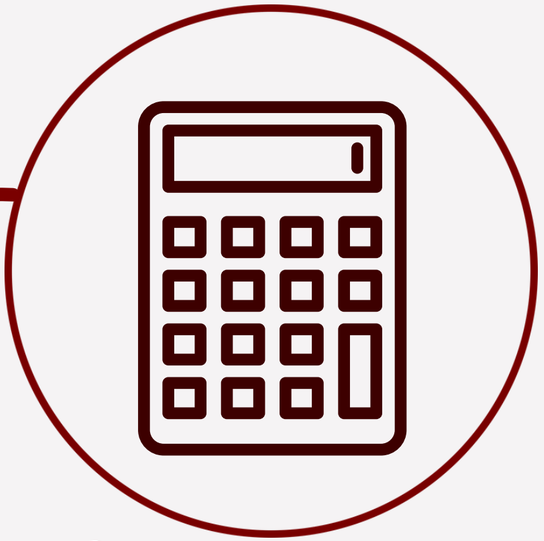
CHOOSING THE RIGHT METHOD

PCF

VS

PCA





FACTOR SELECTION

3 FACTORS

THE ONLY SOLUTION THAT GIVES MEANINGFUL FOOTBALL PROFILES

KAISER'S CRITERION

→ **4 FACTORS**

PEARSON CRITERION

→ **3 FACTORS → EXPLAIN 64%**

SCREE PLOT

→ **ELBOW AT 3 OR 4**

WHY WE IGNORED 4TH FACTOR → **FACTOR 4 DEPENDED ONLY ON THE VARIABLE G-XG**

FACTOR INTERPRETATION



OUTLIERS

WE DETECTED SOME OUTLIERS AFTER FACTOR ANALYSIS



TOO GOOD OR TOO WEIRD → OUT



ALIENS DETECTED

BIG 5

- LINDSTRØM
- HONORAT
- MARMOUSH
- SCHICK
- NZOLA
- DEMBÉLÉ
- LAMINE YAMAL

SECONDARY LEAGUES

- AROKODARE
- ESTÊVÃO
- HEDGES
- LANG
- WEGHORST
- GYÖKERES

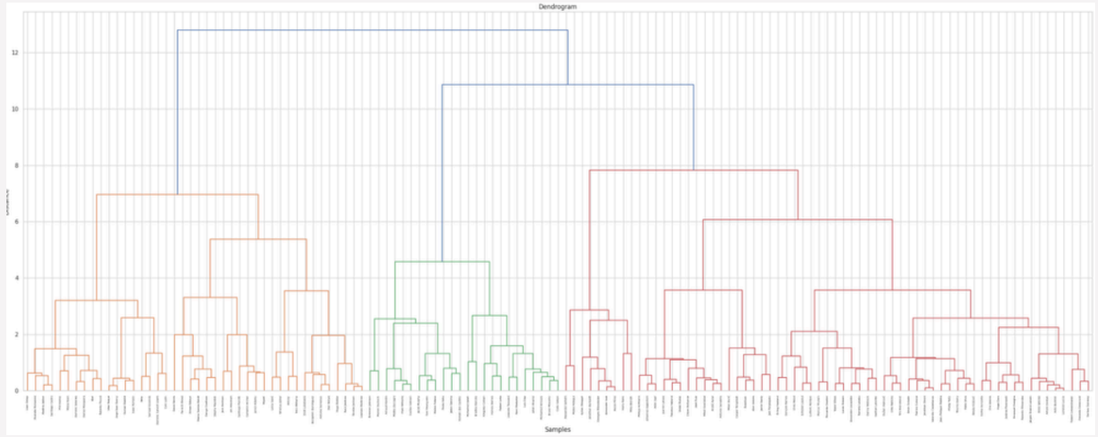
CLUSTERING PROCESS

FIND THE NATURAL GROUPS OF PLAYER PROFILES

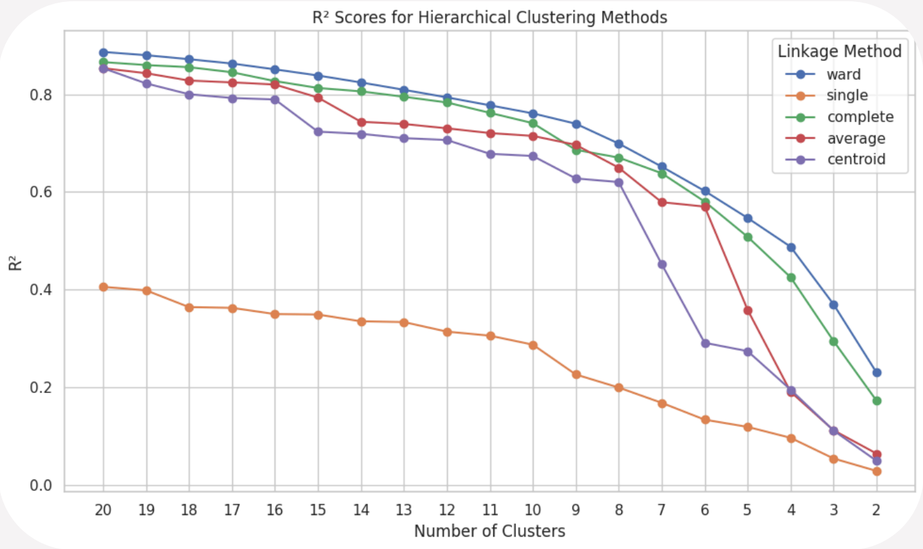
K-MEANS ON 3 FACTOR SCORES

BUILD THE 5 PLAYER ARCHETYPES

HIERARCHICAL CLUSTERING



ELBOW PLOTS



FIVE CLUSTERS FOR BOTH DATASETS

CLUSTER ARCHETYPES



BIG 5

FIVE DISTINCT FORWARD PROFILES

- ELITE PLAYMAKING PLAYERS
- CHAOTIC PRESSING FORWARDS
- SECONDARY/SUPPORT FINISHERS
- LOW IMPACT ATTACKERS
- PURE FINISHERS

CLUSTER ARCHETYPES



SECONDARY LEAGUES

FIVE DISTINCT FORWARD PROFILES

- CREATIVE ENGINES
- ENERGETIC BUT ERRATIC PLAYERS
- UNPREDICTABLE FORWARDS
- LOW-IMPACT ATTACKERS
- PURE GOALSCORERS

MONEYBALL FRAMEWORK

3 RULES TO FIND HIDDEN BARGAINS

Quality

ABOVE CLUSTER AVERAGE

Financial

MARKET VALUE < MEDIAN OF THEIR CLUSTER

Age

YOUNG : ≤23
PRIME : 24–29
VETERAN : ≥30



HIDDEN BARGAINS

★ BIG 5 ★

TOP 5

- **FRANCK HONORAT**



- **PATRIK SCHICK**



- **MIKA BIERETH**



- **GEORGES MIKAUTADZE**



- **ANTONY**



HIDDEN BARGAINS

★ SECONDARY LEAGUES ★

TOP 5

- MOHAMED NASSOH



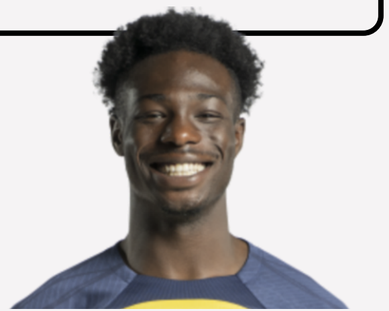
- MITCHELL VAN BERGEN



- WOUT WEGHORST



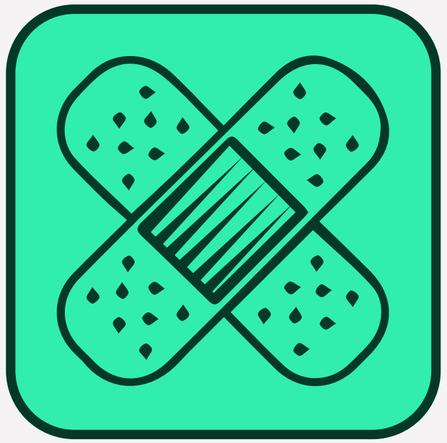
- PROMISE DAVID



- SAYFALLAH LTAIEF



DISCUSSION AND LIMITATIONS



INJURIES



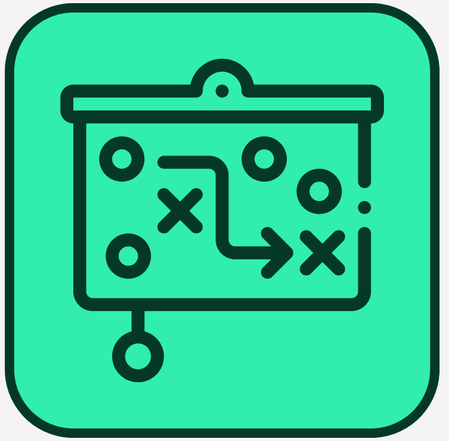
MARKET VALUE



COMPLEXITY



BIAS



TACTICAL BIAS

CONCLUSION

- MULTI-LEAGUE MODELING
- DIMENSIONAL ANALYSIS
- PLAYER ARCHETYPES
- IDENTIFYING UNDERVALUATION
- TARGETING EFFICIENCY

